

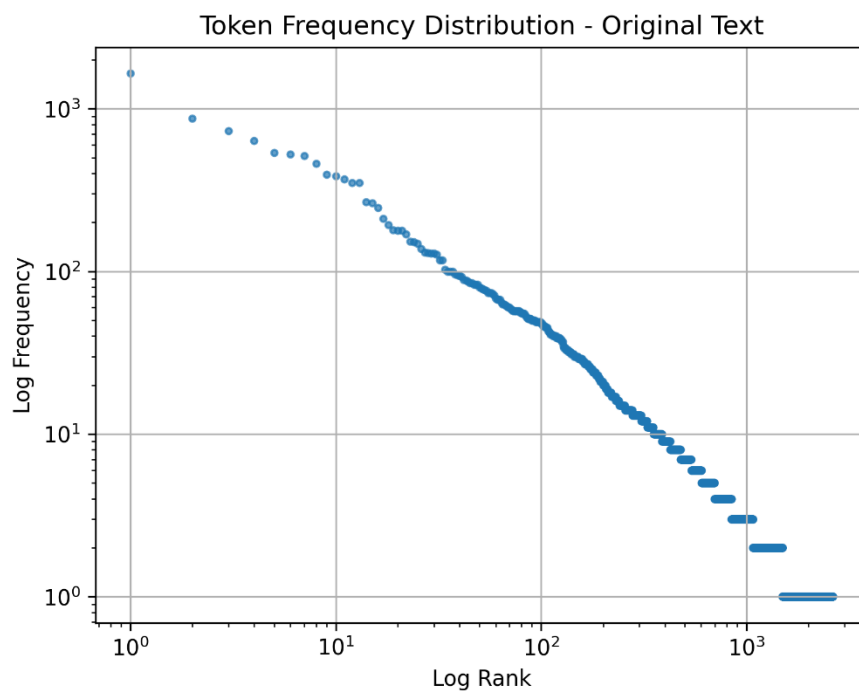
HW3 - 67978

Group #34

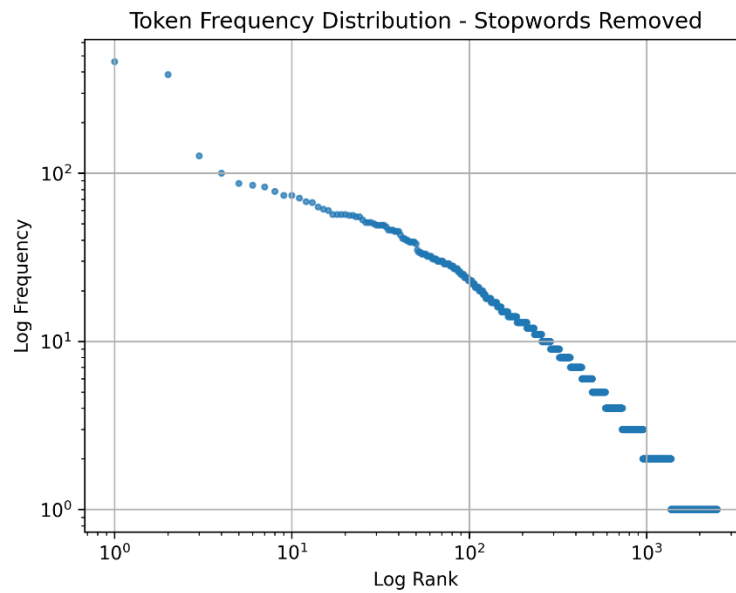
IDs: 208611939, 322336116, 208323345

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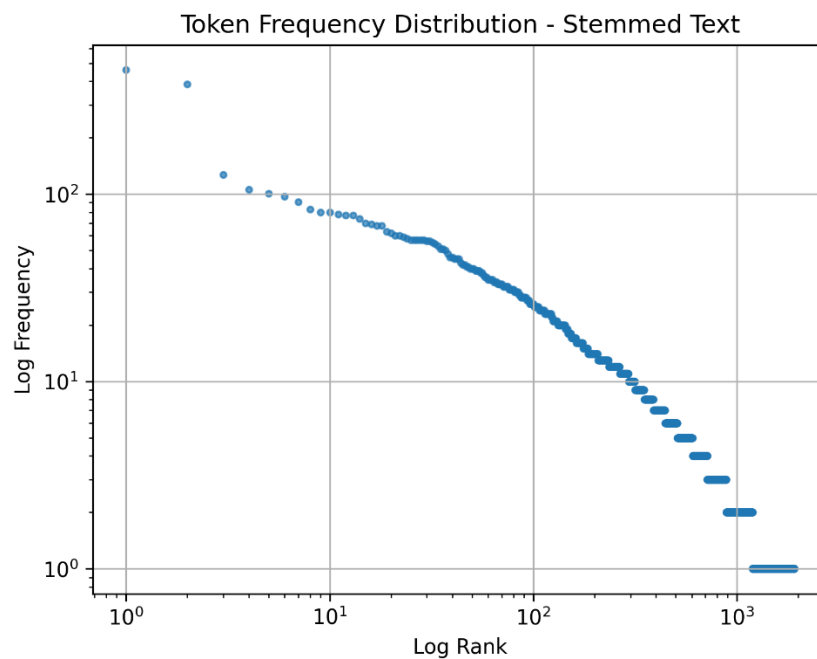
1. The book we chose is "Alice's Adventures in Wonderland", by Lewis Carroll.
2. After tokenizing the text, we received a total of 26,536 tokens, with 2,628 unique tokens.



3. List of top 20 tokens: [the, and, to, a, she, it, of, said, i, alice, in, was, you, that, as, her, at, on, with, all]
4. We tokenized the text again, after removing stopwords, we received a total of 12,788 tokens, with 2,500 unique tokens.
List of top 20 tokens after removing stopwords: [said, alice, little, one, know, like, went, would, thought, could, time, queen, see, well, king, don't, mock, way, began, it's]



5. We tokenized again with stemmed text, we received a total of 12,788 tokens, with 1,910 unique tokens. List of top 20 tokens: [said, alic, little, look, one, like, know, went, thought, thing, would, time, go, could, say, queen, get, see, well, king]



The original token counts are dominated by common words such as the, and, and to. After removing stopwords, these words disappear, making the most frequent tokens more representative of the book's content.

After stemming, different forms of the same word are merged into a single stem, reducing the number of unique tokens while keeping the total number of tokens the same.

6. In the sentence “How brave they’ll all think me at home!”, the POS tagger labels “brave” as NN, meaning noun. This is incorrect because “brave” describes “me” and functions as an adjective; The correct tag should be JJ. The mistake may be caused by the unusual sentence structure and the contraction “they’ll”, which the tokenizer splits awkwardly.
7. The word cloud accurately highlights the main characters of the story, such as Alice, the Queen, the King, the Rabbit, the Hatter, and the Gryphon. A few common words (e.g., "Well", "Come", and "Oh") also appear because the POS tagger occasionally misclassified them as proper nouns. Overall, the word cloud corresponds well to the main characters and themes of Alice's Adventures in Wonderland.

Some repeated phrases (e.g., Twinkle, twinkle and Beautiful, beautiful) appear multiple times because they occur more than once in the text. Also, Soup! Soup shows as a newline separated between them in the book, still consecutive.

Down, down
Down, down
thump! thump
you: you
Dear, dear
there. There
Prizes! Prizes
Pat! Pat

you, you

it, it

No, no

Wow! wow

Wow! wow

it: it

Twinkle, twinkle

Twinkle, twinkle

twinkle, twinkle

Sh! sh

Two. Two

Hush! Hush

you! You

Tut, tut

No, no

Soup!

Soup

Beautiful, beautiful

Beautiful, beautiful

No, no